Diwali Sales Data Analysis Report

1. Key Observations

- **Total revenue** during the Diwali period is **₹106.25 crore**, generated from **27,981 orders**.
- The **average order value (AOV)** is **₹3,797**, which may indicates customers are purchasing multiple items or mid-to-high value products per order.
- Revenue is **not evenly distributed** across customer segments, products, or regions.

Assumption:
All orders and revenue belong to the same Diwali sales period and are calculated using consistent pricing and tax rules.

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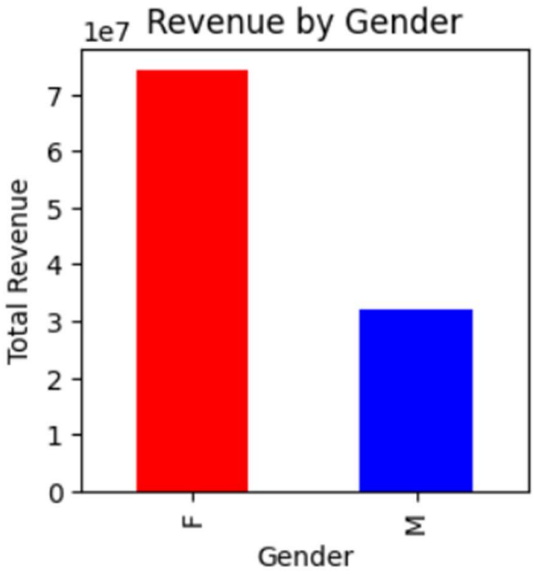
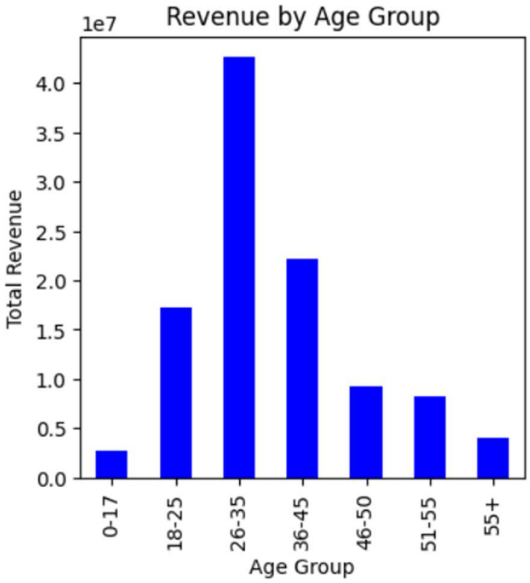
2. Customer Trends





- **Female customers contribute ~70% of total revenue**, while male customers contribute ~30%.
- The **26–35 age group** generates the highest revenue among all age segments.

- **What this means (practically):**
- Female customers are the **primary revenue-driving segment** during Diwali.
- This pattern suggests that marketing, product mix, or promotional messaging may be more aligned with female customers during the Diwali period.
- Customers aged **26–35** appear to have higher purchasing power or higher purchase frequency compared to other age groups.

Assumption:
Gender and age data are self-reported or accurately captured during checkout.

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3. Product Performance

- The **top-performing product categories** by revenue are:
    1. Food
    2. Clothing
    3. Electronics

These categories together account for a significant share of total revenue during the Diwali period.

What this means:

- Customers are prioritizing **consumables and lifestyle products** during Diwali.
- Electronics also perform strongly, suggesting gifting or personal upgrades during the festival period.

Assumption:
Product categories are consistently classified, and revenue is not double-counted across categories.

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4. Geographic Insights

- The **top revenue-generating states** are:
    - Uttar Pradesh
    - Maharashtra
    - Karnataka

Higher performance in these states may also reflect stronger distribution coverage or higher order volumes.

Central and Western zones contribute the majority of revenue.

What this means:

- Sales are **concentrated in a few large states**, likely due to population size, purchasing power, or distribution reach.
- Certain zones clearly outperform others in terms of revenue contribution.

Assumption:
Customer location is determined using accurate delivery or billing address data.

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5. Observations & Considerations

- **Revenue concentration consideration:** Heavy dependence on a single gender group and a few states may limit overall balance.
- **Category reliance:** Strong reliance on a few product categories could impact revenue if demand shifts.

- **Uneven demographic contribution:** Lower contribution from other age groups may indicate under-engagement.

Note:
These are **current observations are based on existing data**, not forecasts or predictions.

6. Data Quality Notes

- A small number of records had missing revenue values and were excluded from calculations.
- The analysis assumes consistent data entry for customer demographics and product categories.
- No external data or assumptions beyond the provided dataset were used.